AIM

To observe the decrease in pressure with the increase in velocity of a fluid.

YOU NEED

1. A strip of paper, 2. Two ping-pong balls, 3. A stand, 4. Thread

THEORY

Bernoulli's theorem states that in the streamline flow of an ideal liquid through a horizontal tube, the velocity increases whereas the pressure decreases, and vice-versa.

HOW TO DO

A. Lifting of a paper strip on blowing:

1. Take a small paper strip and place it in front of your mouth.
2. Now blow the air over the strip of paper in horizontal direction as shown in the figure and observe.

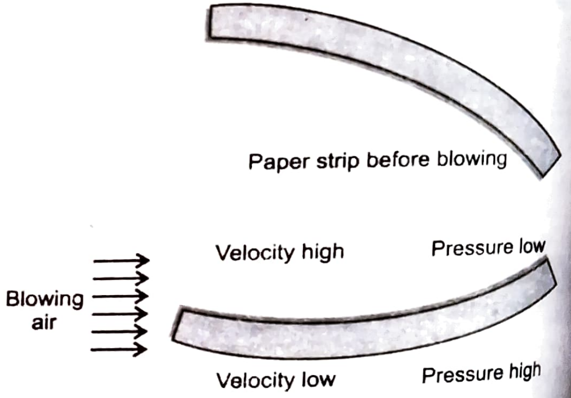

Fig. 41. Lifting of a paper strip on blowing.

OBSERVATION

It is observed that the paper strip moves up.

CONCLUSION

On blowing air over the paper strip, the velocity of air increases, which creates low pressure above the air and high pressure below the air. This difference in pressure, lifts the paper strip.

between two ping-pong balls: